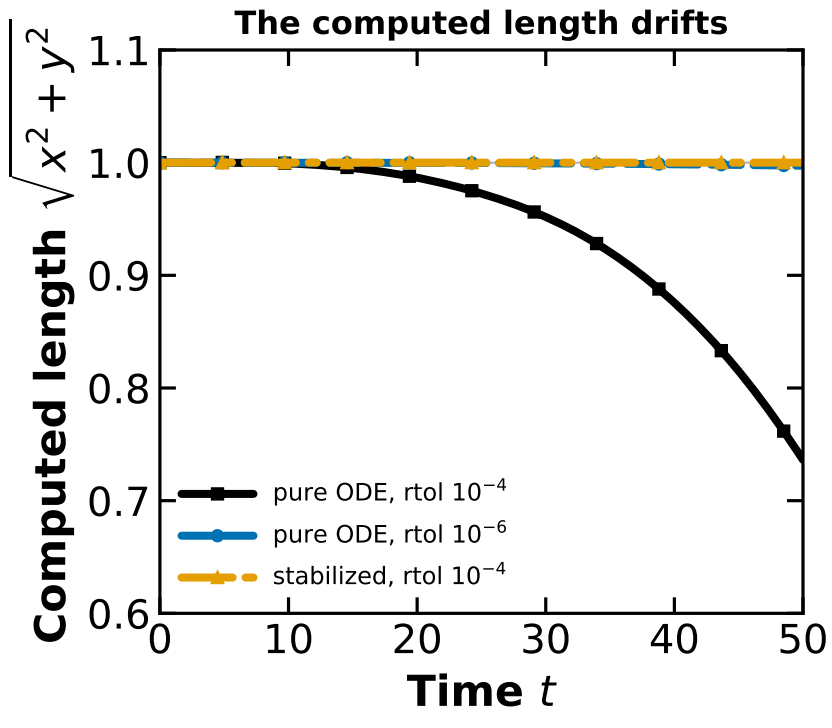
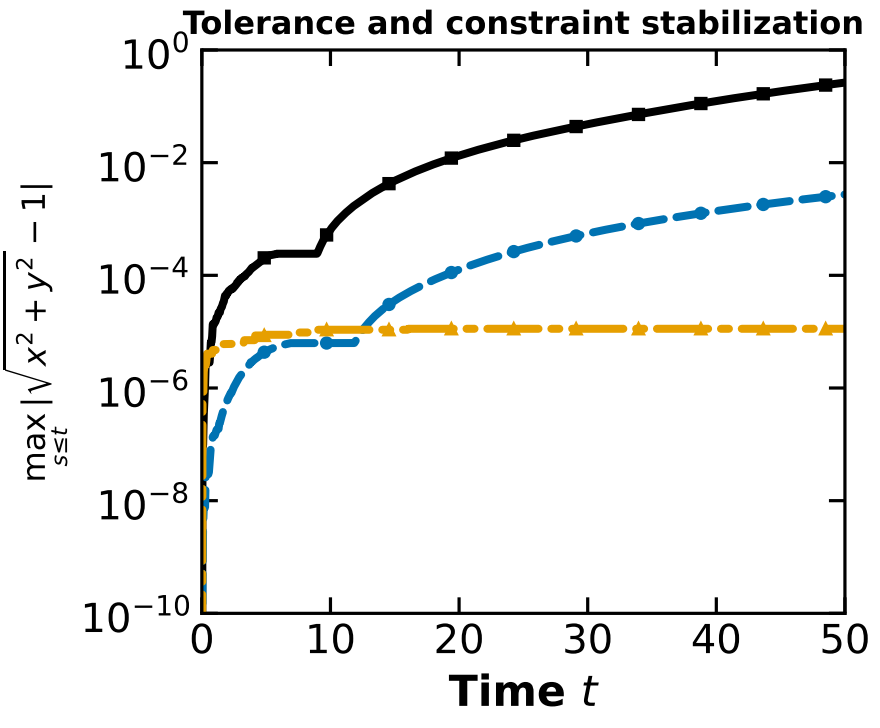

The computed length drifts
Computed length $\sqrt{x^2+y^2}$
1.1
1.0
0.9
0.8
0.7
0.6
0
10
20
30
40
50
Time t
pure ODE, rtol 10^{-4}
pure ODE, rtol 10^{-6}
stabilized, rtol 10^{-4}


Tolerance and constraint stabilization
$\max_{s\le t} |\sqrt{x^2+y^2}-1|$
10^{0}
10^{-2}
10^{-4}
10^{-6}
10^{-8}
10^{-10}
0
10
20
30
40
50
Time t